Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Full-stack AI engineer with 6+ years of experience shipping features in enterprise SaaS at Entrata (100K+ monthly active users). AI tools-native: daily Claude Code and Cursor user. Currently integrating Anthropic Claude into financial applications at Intuit - building internal tools for finance teams, backend APIs, and LLM-powered web UIs. Built two AI projects end-to-end: AgentProbe (15-endpoint FastAPI + 3 LangGraph agents, multi-provider LLM evaluation) and Leasel (8-agent LangGraph workflow, hybrid-search RAG, HITL escalation). Case study: github.com/venkatalakshmiparimala/AgentProbe. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Full Stack Engineering | AI Agents & LangGraph | FastAPI & Backend APIs | React & TypeScript | Internal Tools | Claude Code & Cursor | LLM Evaluation | RAG Pipelines | Python (9+ yrs) | Enterprise SaaS | Production Systems | Cross-functional Delivery

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Build **internal tools for finance and operations teams** - natural-language query interfaces, automated reporting pipelines, and React dashboards that reduce manual workload for non-engineers.
- Integrate Anthropic Claude into Python/FastAPI services for financial data summaries, natural-language queries, and reporting; design prompts and structured-output schemas for consistent, traceable answers.
- Develop backend APIs and data pipelines with React/TypeScript front ends deployed to AWS (S3, EC2, Lambda, API Gateway) via GitHub Actions CI/CD.
- Build LLM evaluation and regression tests (LLM-as-judge and golden datasets) to catch hallucinations and quality drops before release; add guardrails for input validation and sensitive-data handling.
- Build ML pipelines with scikit-learn and XGBoost - data ingestion, feature engineering, model evaluation, and prediction APIs.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant enterprise SaaS, 100K+ monthly active users

- Delivered **20+ features end-to-end** (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints, owning design through deployment with cross-functional teams.
- Built **internal ops tools** (Python/Bash) for operations and finance teams - eliminated 80% of manual workload and cut incident triage time by 40%.
- Built a RabbitMQ-based async processing system for rent updates across thousands of tenant records, with retry logic and idempotency guarantees.
- Optimized 200+ SQL queries using execution plans and indexing, improving performance by 30% across multi-tenant production databases.
- Maintained the yearly HUD income-limit import for 7 years - loading federal data into ~180,000 records with validation and rollback each cycle; zero compliance violations.
- Docker microservices redesign (50% better scalability, 30% faster deployments); AWS S3 migration with Libsodium encryption for 100% PHI compliance.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built property management modules (Python, PHP, Java, PostgreSQL) for affordable housing and rental management.
- Delivered large-scale data import and reconciliation workflows (1.6M HUD/Tax Credit records); led AWS S3 cloud migration from legacy server storage.

PROJECTS

AgentProbe - Open-Source LLM Evaluation Toolkit Live on GitHub

Built a 15-endpoint FastAPI backend routing prompts across five LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) via a smart LLM Router. Scores LLM and RAG outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. Three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) run 100% locally at zero cost. CI regression tests with golden datasets across 1,240+ test cases.

Python · FastAPI · LangGraph · ChromaDB · LangChain · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama · GitHub Actions - github.com/venkatalakshmiparimala/AgentProbe

Leasel - Multi-Agent AI Platform with RAG

Built an 8-agent LangGraph workflow (triage, retrieval, reasoning, validation) that investigates support tickets and escalates to humans when confidence is low. RAG pipeline: OCR ingestion, section-aware chunking, hybrid search (TF-IDF + semantic reranking), cited answers. Guardrails enforce that eligibility and financial decisions always route to staff, never the LLM. LLM-as-judge CI harness: 93.6% precision across 1,240 test cases; 100% citation correctness on Phase 1 eval set.

Python · LangGraph (8-agent) · FastAPI · Streamlit · TF-IDF RAG · OCR · SQLite · CI/CD - github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR - Clinical NLP & LLM API Integration

Built a full-stack app translating English/Spanish questions into FHIR API calls using LLMs, BERT, and embeddings - cut manual clinical lookup time by ~60%. React/TypeScript frontend, Flask backend, Docker, HIPAA-style RBAC.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs - github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University May 2026

GPA 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — JNTU Anantapur 2017

TECHNICAL SKILLS

AI & LLMs: Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, AI Agents, LangGraph, LangChain, HITL, LLM-as-Judge, Claude Code, Cursor, Codex, MCP

Vector Search & NLP: ChromaDB, Pinecone, Weaviate, Elasticsearch, Semantic Search, BERT, spaCy, HuggingFace

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark

Web & Other Languages: TypeScript, JavaScript, React, Node.js, Java, Spring Boot, PHP, Laravel, GraphQL, SQL, PL/SQL

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Practices: REST APIs, Microservices, Agile/Scrum, TDD, Code Review, HIPAA/PHI Compliance, Observability, Internal Tools